Chapter 11
Innovation and Social Impact in Software Engineering:
A Practical Experience From the Fábrica de Software

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ABSTRACT

This study addresses the importance of software engineering in society, highlighting the need for a practical approach in university education through a concise training center that currently plays a role in the education of its members. From this perspective, an initiative of the Federal University of Pará, the Fábrica de Software (FS), is presented, where the application of agile and structured methodologies and the frequent support of the faculty are emphasized, with the aim of improving and developing the professional side of the involved students. In this research, the methodological construction used to guide the FS participants will be detailed, while the concrete developments achieved by the Factory will be presented. This methodology encompasses different spheres of everyday life, such as health, education, and culture, and includes the initiative's commitment to the community, showing how knowledge is transmitted and the socio-educational role of the FS.

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INTRODUCTION

Software Engineering, recognized for its systematic and disciplined approach towards the development, maintenance, and operation of software systems, plays a pivotal role in contemporary technological advancements (Tsui, F., et al., 2022). This discipline is crucial in driving innovations in critical areas such as health (Sendak, Mark, 2023), education (L. van der Duim., et al., 2007), and the industry (Garousi, Vahid, et al., 2019), significantly shaping society and everyday life. Josephine Wolff (2021) highlights the capability of Software Engineering to simplify and enrich daily tasks through technological advancements, transforming routine activities with innovative solutions that improve efficiency and accessibility in essential services.

While the integration of Software Engineering practices into society is broadly advantageous, recent research, such as by Tsui, F. (2022), points to a significant variation across different areas of society, revealing a stark discrepancy between the potential of these technologies and their actual implementation across various sectors. This disparity highlights a central issue: while some fields, such as the IT sector, rapidly advance in integrating these practices, others remain significantly behind, limiting the reach and benefits of technological innovations. Among these challenges are the proper allocation of resources, resistance to change, integration of these initiatives with existing curricula, and the sustainability of adopted practices. Furthermore, issues related to infrastructure, access to relevant technologies, and adequate training for educators also emerge as obstacles to be overcome. Yingxu (2021) also emphasizes the importance of reconfiguring our modes of communication, work, and leisure through mobile platforms, pointing to the need to expand these innovations to benefit a broader range of domains.

In this sense, driving the adoption of Software Engineering practices in society is crucial to fostering innovation, well-being, and various beneficial solutions for society. By integrating these practices in conjunction with the educational curriculum, future professionals are prepared to face contemporary challenges with effective technological solutions, promoting a culture of continuous improvement and critical thinking (Kelley and Knowles, 2016). This approach not only enriches the professional development of individuals, preparing them for the demands of the job market, but also ensures that society as a whole benefits from the advantages that technology can offer, from process optimization to creatively and efficiently solving complex problems (Eppinger, E., 2021).

In this context, Software Engineering emerges as an essential pillar in the development of systems that significantly impact various aspects of everyday life. The effective transmission of this knowledge, especially in universities, becomes imperative, requiring a practical and contextualized approach (Pacola, Vinicius, 2021). By mastering practical skills in this field, students from higher education

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